

WINDOWS PRINTING ARCHITECTURE AND PROCESS FLOW

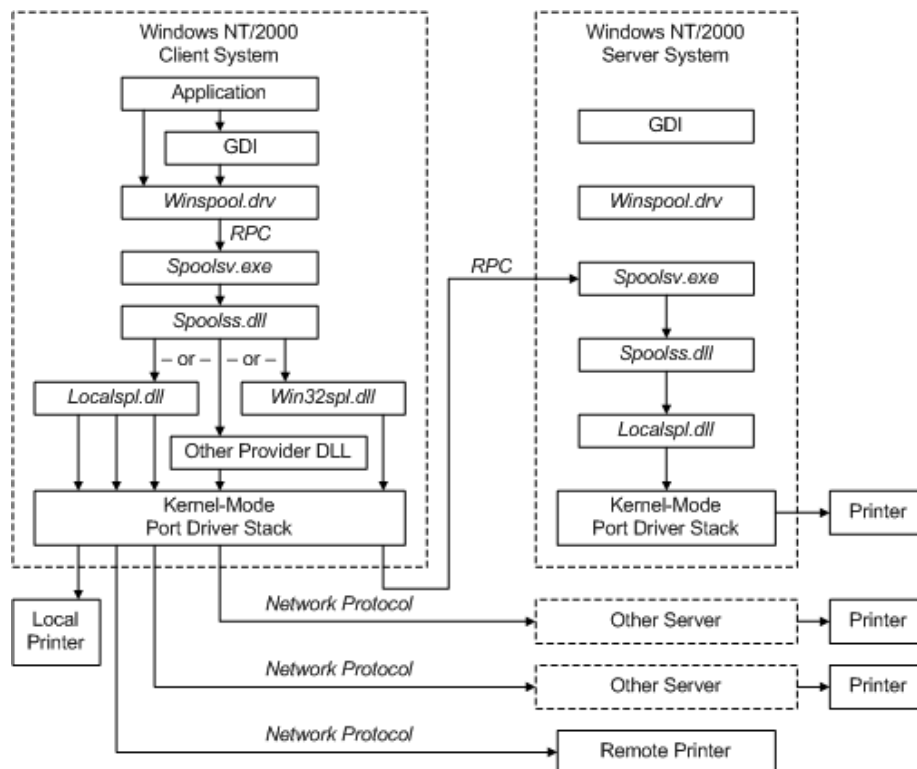
A Print Server is defined as a printer attached to a local computer that is shared over a network to facilitate centralized printing services. The specific configuration of a computer sharing its local printer over a network characterizes that computer as a Print Server.

In the context of the Windows operating system, the term **Windows Printer** refers to a comprehensive system that enables software applications to produce hard copy output on a physical printing device. This process involves the conversion of digital data into a printable format, a task that is initially performed as an application generates a print job. The operating system is responsible for the systematic management of these print jobs, which includes the queuing of documents, the processing of data streams, and the subsequent transmission of data to a designated printer.

The Windows printing architecture is strategically designed to allow applications to print documents or jobs to a wide variety of printing devices. This architecture supports two primary classifications of printing:

Local Printing and Network Printing. Local Printing involves the process of printing to a device that is directly connected to the client machine. A local printer is typically connected via physical or wireless interfaces such as USB, ETH, or WiFi.

The technical flow of a print job within the Windows environment is structured as follows:



When a user issues a print command from any application, the nature of the data determines the initial path. If the data consists strictly of plain text, it is directed to Winspool.drv, which is formally identified as the client spooler interface. However, if the data to be printed contains graphical elements, the request is first sent to the Graphical Device Interface (GDI). The GDI converts the complex graphical data into a standardized form that is compatible with Winspool.drv.

Upon completion of the conversion, the request is routed to Winspool.drv. Functioning as the primary client interface for the spooler, it invokes the Spooler Service, identified as Spoolsv.exe. The Spooler Service is a fundamental component of the Windows operating system that manages all active print jobs.

Network Printing involves the transmission of print data to a printer shared across a network infrastructure. In this scenario, the print job is dispatched from a client machine to a designated Print Server. This Print Server may be another client machine or a dedicated server device. This architecture enables multiple users to share a single printing resource effectively.

The printing process begins in earnest when the Spooler Service calls the Router, implemented as spoolss.dll. The Router performs the critical function of routing the print request to a specific and appropriate print provider. The Router determines which print provider to invoke based on the printer name or the handle provided in the job request. Once the determination is made, the Router calls the required print provider to manage the document flow.

Print Providers are implemented as dynamic link libraries (dlls). These components are generally categorized into two primary types:

1. localspl.dll: This component is responsible for managing print jobs for printers that are connected locally to the system.
2. Win32spl.dll: This component is utilized to handle print jobs for printers situated on Windows network Servers.

Print providers ensure that print jobs are directed to their correct destinations, whether those destinations are local hardware devices or network-accessible printers.

Following the routing stage, the print job is directed to the Print Processor for final data transformation. The Print Processor stage is where the critical conversion from Enhanced Metafile (EMF) format to the Raw format occurs. The default print processor utilized within the Windows operating system is known as **WINPRINT**.

The Print Processor is responsible for converting the spooled data of a print job into a specific format that a print monitor can transmit to the physical printer. This conversion is essential because software applications generate print data in various formats, whereas physical printers require specific, standardized data formats to execute the print task.

In addition to data conversion, the Print Processor reads the spool file and performs necessary conversion operations on the data stream before writing the converted data back to the spooler. This component also handles application-level requests to manage the job state, such as pausing, resuming, or canceling active print jobs. After the conversion process is finalized, the Print Processor transmits the formatted data to the Print Monitor. The Print Monitor conducts checks on the available ports to assign the printer, ensuring the hardware receives the data correctly for physical output.

Print Monitor: The Print Monitor is the architectural component responsible for assigning print jobs to the specific port on which the printer is currently connected. The processed print job data is systematically sent from the Print Spooler to the printer. The system ensures that the data is transmitted directly, correctly, and reliably to the printer hardware. The Printer serves as the final destination and is the physical device that provides the hardcopy output to the user. -> In the context of Network Printing, the router calls the Win32spl.dll provider, from where the print job is directed to the spooler service (spoolsv.exe) of the specific machine which has the printer connected to it locally. From that point, the print job processes in the same manner as it does in the case of local printing. The print job is directed to the spooler service over port TCP-135 (RPC).

Spooler Components.

Application -> The Print application initiates the process and creates a print job by calling a GDI function. The Graphical Device Interface (GDI) is used by applications that require extensive graphic support or specialized data conversion.

Winspool.drv -> This component functions as a client interface into the spooler subsystem. This is a driver file primarily responsible for calling the spooler service, which is identified as spoolsv.exe.

Spoolsv.exe -> This component is implemented as a system service that is automatically initialized and started when the operating system is started.

Spoolss.dll -> This library acts as a router, determining which specific print provider to call based on the printer name or handle supplied with each specific function call.

Print Provider -> The print provider is the component responsible for directing print jobs to either a local or remote print device and managing the associated print queue. Print provider is of two types.

1. **Local spl.dll** -> This is a local print provider which handles all print jobs that are directed specifically to locally connected printers.
2. **Win32spl.dll** -> All print jobs that are directed to a Remote Server are managed and handled by this provider.

Print Processor -> These components are designed to convert spooled data from a print job into a data format that is correctly understood and processed by the print monitor. The default print processor provided with the operating system is WINPRINT.

Print Monitors -> This component is responsible for directing the print data stream from the spooler to the appropriate port drivers for final transmission. It is also responsible for the systematic management and technical configuration of printer ports on a server.

PRINT DRIVER Print drivers are software components that translate Windows output data into a printer control language that a physical printer can interpret and understand. Print drivers are typically provided by printer manufacturers and are developed specifically based on the hardware model of the printer.

Print drivers are categorized into two primary types: **Type-3 (Legacy)** and **Type-4 (modern)**.

Type-3: These are Vendor Provided Drivers that are also maintained and updated directly by the hardware vendor.

Type-4: This driver type comes bundled with the Windows operating system and is updated automatically through the Windows Update service.

Print driver isolation: Print driver isolation is a mechanism that allows administrators to troubleshoot performance or stability issues with the printer driver by isolating the printer driver within a separate container environment. The associated executable for this process is PRINTISOLATIONHOST.EXE.

-> System administrators can Enable branch office direct printing to optimize network traffic and printing efficiency in remote locations.

-> It is important to define the RAW and EMF (Enhanced Metafile) datatypes and their roles within the printing subsystem.

-> Understanding the technical process of conversion from RAW to EMF and vice-versa is critical for managing print data streams.

There are three distinct kinds of **Driver isolation** available in the system configuration.

(i) None: There is no isolation implemented in this specific scenario.

(ii) Shared: In this configuration, multiple print drivers can be isolated together within a single shared container process.

(iii) Isolated: In this configuration, a specific print driver is isolated individually in its own separate and dedicated container.

What is client side rendering: (CSR) Client side rendering is a configuration setting which tells the system that print jobs should render on the client computer rather than the server.

Server Side Rendering: In this operational mode, the print jobs are rendered and processed on the print server.

What is Point and Print. ->

Point and Print is a specialized method used for installing printers on client systems. In this method, the system will check if there is a compatible driver available in the local spooler repository. If a compatible driver is present, the system will install it directly from the local repository; if it is not present, the system will automatically download the necessary driver from the print server.

What is Print Nightmare.

Print Nightmare refers to the critical security vulnerabilities identified in the Windows Print Spooler service that could allow for remote code execution.

Print Nightmare is a known vulnerability in the Windows print spooler service which allowed attackers to execute code remotely and leverage privileges locally. To patch this critical vulnerability, Microsoft released a series of updates starting on July 6, 2021, which introduced the registry key `RESTRICTDRIVERINSTALLATIONTOADMINISTRATOR`, abbreviated as `RDITA`. This value was initially defaulted to '0' (disabled) until Microsoft released a subsequent update on August 10, 2021, where the value was defaulted to '1' (enabled).

This registry setting, found at the path `HKLM\SOFTWARE\Policies\Microsoft\WindowsNT\Printers\PointAndPrint`, restricts standard users from installing printer drivers from print servers.

Furthermore, the executable `PRINTBRM.EXE` is responsible for the migration of printers from one system to another.

Initial communication between the print server and the client was conducted on port 445 (SMB), but this communication now occurs over port TCP-135 (RPC).

Section VI: Troubleshooting and Technical Specifications

The '11B' error is a common issue where both the client and server Windows installations need to be on the same patch or Windows update level; if they are not synchronized, users are likely to encounter this error when attempting to add printers. If the update levels are confirmed to be correct and the issue persists, it could indicate a network error or that the required port range is being blocked.

Key acronyms in this domain include

`RDITA` for Restrict Driver Installation To Administrator,

XPS for XML Paper Specification, and

RALPE for RPC Packet. Additionally,

RPC Authn Level Privacy is typically Enabled (1) for secure communication.

The Enhanced Metafile, or EMF, is an image description format defined by Microsoft that describes the appearance of text, graphical shapes, and sampled images on printed or displayed pages. RD Easy Print, which was previously known as TS Easy Print, is utilized for redirecting printers on a remote computer. For the redirected printer on the remote system, the driver is used to send the print jobs back to the locally connected printer.

A printer driver includes two primary functional components: the rendering component and the configuration component.

The Rendering Component converts graphics commands from the software application into a specific format that the printer hardware understands.

The Configuration Component contains the user interface that allows users to control printer-selectable options and includes the program interface that communicates the printer's specific configuration and features to an application.

Enhanced Point & Print refers to the updated printer sharing mechanism that allows a print client to print to V4 shares without the necessity of downloading the manufacturer-provided device driver from the Print Server.

The Windows printing subsystem supports two generations of drivers with distinct characteristics:

Type-3 / V-3 Drivers are considered Legacy components. They are provided directly by printer manufacturers and vendors.

Printer management for Type-3 is notably complex, requiring separate 32-bit and 64-bit drivers as well as different drivers for different hardware models.

These drivers take more disk space due to the inclusion of custom configuration UI and language resources.

Because they are developed independently by manufacturers, V3 drivers are historically more prone to system crashes. Updates for these drivers are provided solely by the manufacturer.

Type-4 Drivers represent the Modern architecture, often referred to as Print Class Drivers. These are bundled with the operating system and updated through Windows Update.

Microsoft works closely with IHVS (Independent Hardware Vendors) for the development of V4 drivers. Printer management is significantly simplified as a wide range of printers are supported through common PDLs (Page Description Languages) such as PCL, PS, and XPS.

They utilize smaller, Windows-provided components and are less prone to crashes due to their reduced size and modular nature. Updates are pushed seamlessly through Windows Update.

There are various methods available to add a printer from a Print Server using Point & Print:

1. Adding from the Printers and Scanners section within the Windows Settings menu.
2. Adding via a UNC path.
3. Utilizing the Drag and Drop method.
4. Through a right-click selection.

POINT AND PRINT AND SYSTEM COMPONENTS

What is the Point & Print feature? It is a comprehensive feature that allows users to print to a remote printer on a network without having to manually download and install the printer driver on the machine.

The Point & Print mechanism checks for an existing driver (printer driver) on a local driver store, and if it does not find a compatible driver, it downloads and installs the driver from the print server itself.

- Client downloads the configuration data from the server using GetPrinterDataEx calls when using V4 drivers.
- The configuration data includes essential data files such as Generic Printer Description (GPD) files, PostScript Printer Description (PPD) files, and resource DLLs (RC.dll), etc.
- The print systems examine the client and validate that the resource DLL (RC.dll) contains no executable code to ensure system security.
- Splwow64.exe: This is a system process that allows 32-bit applications running on 64-bit Windows environments to interact successfully with 64-bit printer drivers.

PRINT PIPELINE AND NETWORK PROTOCOLS

LPT -> Line Print Terminal (Parallel Port)

Print Filter Pipeline: The Print Filter Pipeline is a component in XPSdrv print drivers that processes the XPS spool file and includes the processing filters and the filter configuration files. It represents the logical collection of printer driver filters that are specified in the filter configuration file.

Terminal Services Printing (TS Easy Print) and Printer Redirection.

Filter Pipeline:

- i) Specifies filters to load
- ii) Processes jobs

iii) Unloads filters

iv) Exits on process ends or another event.

Filter Pipeline Configuration Services: PrintFilterPipelineSvc.exe

WSD: Web Services on Devices is a specialized protocol for communicating with a device (printer) over the network. WSD allows network-connected IP-based devices to advertise their functionality and offer these services to clients by using the Web Services Protocol. WSD provides a network PnP (Plug and Play) experience that is similar to installing a USB device.

Hardware ID: It is a unique identifier assigned by the enumerator associated with the port that the printer is connected to. A Hardware ID is a vendor-defined identification string that Windows utilizes to match a device to a specific inf file. A device can have more than one hardware ID associated with it.

DRIVERLESS PRINTING AND MOPRIA

Driverless Printer -> MOPRIA model MOPRIA Certified Printers: When a dedicated driver is not available, Windows can install MOPRIA certified printers without needing to install any additional software or drivers. What is MOPRIA? -> MOPRIA refers to the Mobile Printer Alliance.

Mopria is an alliance of printer manufacturers that are working together to define an Open Standard for Driverless Printing that can be used to make printing from mobile devices significantly easier. It is an implementation of the IPP (Internet Printing Protocol) Standard.

REGISTRY KEYS AND RENDERING MODES

Important Reg.

- HKCU \ Software \ MS \ WINNT \ CurrentVersion \ Devices -> This key contains the list of all printers installed on the system.
- HKLM \ SYSTEM \ CurrentControlSet \ Control \ Print \ Environments \ Printers

Q. What is Client Side Rendering? -> It is a feature that allows print jobs to be fully rendered on the client machine when targeting a shared printer hosted on a Windows client or server computer acting as a print server. After being rendered on the client, the print job is sent to the print server for queuing and printing by the print spooler without further server-side rendering required.

Q. When printing through the print server instead of the job being rendered on the client, the job is sent to the server where the job gets rendered and subsequently sent to the print spooler for queuing and printing.

- Type-3 driver location in file The Type-3 printer driver location within the file system architecture is found at the specific path C:\Win\Sys32\Spool\drivers\x64\3.

Printer Redirection.

Printer Redirection is a specialized feature that enables a locally connected printer to be mapped onto a remote machine, allowing for printing capabilities across a network or over the internet. This functionality specifically enables the user to output documents to their locally installed printer while connected to a Terminal Server Session, which is formally identified as a Remote Desktop Session. The RDS client provides the underlying framework required to facilitate this redirection of data between the local hardware and the remote session environment.

WINDOWS PRINTING TECHNICAL TOPICS AND STUDY QUESTIONS

SECTION 1: FUNDAMENTALS OF WINDOWS PRINTING

- Definition of a Print Server and shared local printers.
- Overview of the Windows Printer system and hardcopy output.
- Classifications of printing: Local Printing (USB, Ethernet, WiFi) and Network Printing.

SECTION 2: THE WINDOWS PRINT SPOOLER ARCHITECTURE

- Data routing paths: GDI for graphics versus Winspool.drvc for plain text.
- The role of Winspool.drvc as the primary client interface.
- Technical functions of the Spooler Service (spoolsv.exe).
- Routing logic performed by Spoolss.dll.

SECTION 3: PRINT PROVIDERS AND NETWORK ROUTING

- Functions of Local Print Providers (localspl.dll).
- Functions of Network Print Providers (Win32spl.dll).
- Network communication protocols: Transition from Port 445 (SMB) to Port 135 (RPC).

SECTION 4: DATA TRANSFORMATION AND OUTPUT MANAGEMENT

- Print Processor operations: Conversion from Enhanced Metafile (EMF) to RAW format.
- Characteristics of the default WINPRINT processor.
- Print Monitor responsibilities: Port assignment and hardware communication.

SECTION 5: PRINTER DRIVER ARCHITECTURE (V3 VERSUS V4)

- Characteristics of Type-3 (Legacy) Drivers: Vendor provided, complex management, and 32/64-bit requirements.
- Characteristics of Type-4 (Modern) Drivers: Bundled with OS, Windows Update delivery, and Print Class Driver architecture.

- Functional components: Rendering Component versus Configuration Component.

SECTION 6: PRINT DRIVER ISOLATION AND STABILITY

- Purpose of Print Driver Isolation in troubleshooting and system stability.
- The role of PrintIsolationHost.exe.
- Isolation configurations: None, Shared, and Isolated.

SECTION 7: SECURITY VULNERABILITIES AND PATCH MANAGEMENT

- Analysis of the PrintNightmare vulnerability and remote code execution.
- The RDITA (Restrict Driver Installation To Administrator) registry key and its default values.
- Troubleshooting the 11B error and synchronization of Windows Update levels.
- Registry paths for Point and Print restrictions: HKLM
\SOFTWARE\Policies\Microsoft\WindowsNT\Printers\PointAndPrint.

SECTION 8: RENDERING MODES AND REMOTE DESKTOP SERVICES

- Comparison of Client-Side Rendering (CSR) versus Server-Side Rendering (SSR).
- Printer Redirection features in Remote Desktop Services (RDS).
- The role of RD Easy Print (formerly TS Easy Print) in remote sessions.
- Use of Splwow64.exe for 32-bit application compatibility in 64-bit environments.

SECTION 9: DEVICE DISCOVERY AND DRIVERLESS PRINTING

- Web Services on Devices (WSD) protocol and network Plug and Play.
- Mopria Alliance standards and the Internet Printing Protocol (IPP).
- Driverless printing for Mopria-certified hardware.
- Hardware ID strings and vendor-defined identification.

SECTION 10: REGISTRY STRUCTURE AND FILE LOCATIONS

- Key registry paths for installed devices and print environments.
- File system directory for Type-3 drivers: C:\Win\Sys32\Spool\drivers\x64\3.

IMPORTANT STUDY AND REVIEW QUESTIONS

1. Explain the specific scenario where a print job is routed through the GDI versus being sent directly to Winspool.drv.
2. What is the technical responsibility of the Router (spoolss.dll) when a print job is initiated?
3. Describe the two primary types of Print Providers and clarify which one is used for remote server printing.

4. How do Type-4 drivers improve system stability compared to the legacy Type-3 driver architecture?
5. Identify the executable responsible for hosting isolated printer drivers and explain why isolation is used.
6. What are the three levels of driver isolation and how do they differ in their containerization of drivers?
7. Regarding the PrintNightmare vulnerability, what does the RDITA registry key accomplish when the value is set to '1'?
8. Explain the difference between Client-Side Rendering and Server-Side Rendering in terms of where the processing load occurs.
9. What is the function of the Splwow64.exe process in a 64-bit Windows operating system?
10. Define the Point and Print feature and explain how it handles a situation where a compatible driver is missing from the local repository.
11. How does the WSD protocol facilitate the installation of network printers compared to traditional methods?
12. What is the significance of the Mopria certification for modern mobile and driverless printing?
13. Describe the role of RD Easy Print in a Remote Desktop Services environment.
14. Why is it critical for both the client and server to have matching Windows Update levels to avoid the 11B error?
15. What technical conversion takes place within the Print Processor stage of the printing architecture?

Print and Document Services. [Link](#)

Print Services role

What's New in Print Management in Win 7.

Print Scan & Fax Server Installation Guide

Print Management - Step by Step

Assigning Delegated Print Administrator & Printer

Windows Printing Legacy

work with Enhanced Point & Print.